Presentation Script — Mid-Review

Deepfake Detection: XceptionNet vs. CNN-ViT Hybrid on FaceForensics++ · 23AID205 · Target: under 15:00, budgeted at 10:15

#	Presenter	Slides	Starts	Budget
1	Rithvik Arulprakash	1–3	0:00	3:15
2	Vipin Sudhakar	4–5	3:15	2:30
3	Venugopalan Gangadharan	6–7	5:45	2:45
4	Harshith Kv	8–9	8:30	1:45

Rithvik Arulprakash

Starts at 0:00 — 3:15 budget

Slide 1 Title

0:30

• Good [morning/afternoon]. We're presenting our mid-review update for our AIML course project, 23AID205: Deepfake Detection -- comparing XceptionNet against a CNN-ViT hybrid on FaceForensics++.

• Today we'll cover our methodology, our completed data pipeline, and our first experiment's results. Cross-manipulation generalization, compression robustness, interpretability, and the live demo are all underway and will be presented in full at the final review.

• I'm Rithvik -- I'll open with the motivation and our methodology, then hand off to my teammates.

Slide 2 Motivation, Problem & Objectives

1:30

• Deepfakes have moved from a research curiosity to a real misinformation and fraud vector.

• The problem: published detectors usually report one headline accuracy number, on the same manipulation type and compression level they trained on. That number is misleading -- a detector at 99% in a paper can drop to ~57% AUC, worse than random, the moment the test distribution shifts. And that failure is invisible until deployment.

• So our objectives were threefold: fairly compare XceptionNet against a CNN-ViT hybrid -- does adding global self-attention actually help generalization? Measure not just accuracy but full cross-manipulation generalization and compression robustness. And explain *why* generalization succeeds or fails, via Grad-CAM, rather than just reporting a number.

Slide 3 Methodology & Pipeline

1:15

• Our pipeline in six steps: raw FF++ video, MTCNN face extraction, official video-level splits, training, evaluation, and analysis.

• As of today, steps 1 through 3 -- the entire data pipeline -- are complete, and we've carried steps 4 and 5 through for our first experiment. The remaining training and analysis work continues for Experiments 2 and 3.

Handoff: "Vipin will now walk you through our dataset work, which is fully complete."

Vipin Sudhakar

Starts at 3:15 — 2:30 budget

Slide 4 Dataset & Preprocessing — Complete

1:30

• FaceForensics++: 5,000 videos total -- 1,000 real YouTube interviews plus 4,000 manipulated, 1,000 each across four methods: Deepfakes, Face2Face, FaceSwap, and NeuralTextures, at c23 compression.

• We used the official video-level train/val/test splits throughout -- not random frame-level splits, which would leak the same identity into both train and test and quietly inflate accuracy.

• Preprocessing: MTCNN face detection on GPU, 20 frames per video, a 1.3x crop margin to capture blending-boundary artifacts. This gave us roughly 99,987 face crops -- the entire dataset is preprocessed and ready. This stage is fully done.

Slide 5 EDA: Data Quality Check (IQR)

1:00

• Before committing to full preprocessing, we ran an outlier check on every video's frame count and file size using the IQR method.

• [Point to chart] 209 videos flagged on frame count, 326 on file size -- 502 unique outliers overall, about 10% of the dataset -- so data quality was verified upfront, not assumed.

Handoff: "Venu will now introduce our two architectures and our first result."

Venugopalan Gangadharan

Starts at 5:45 — 2:45 budget

Slide 6 Models: XceptionNet vs. CNN-ViT Hybrid

1:30

• XceptionNet is our baseline -- a standard depthwise-separable CNN, ImageNet-pretrained, purely convolutional with only local receptive fields.

• The CNN-ViT hybrid pairs a ResNet-50 feature extractor with a 6-layer, 8-head Transformer encoder. The question we're testing: does global self-attention let the model relate distant regions -- like blending inconsistencies -- that a local CNN structurally can't see jointly?

• Both models share the exact same optimizer, schedule, batch size, and data -- architecture is the only variable, by design.

Slide 7 Results — Experiment 1: Baseline Comparison

1:15

• This is our first completed experiment: both models trained and tested on all four methods together, the same-distribution headline number.

• XceptionNet: 97.29% video accuracy, 0.9958 AUC. CNN-ViT hybrid: 94.71% video accuracy, 0.9836 AUC -- XceptionNet leads on every metric here.

• Whether that holds up under cross-manipulation and compression shift is exactly what Experiments 2 and 3 will show at the final review.

Handoff: "Harshith will close with where we stand and what's coming next."

Harshith Kv

Starts at 8:30 — 1:45 budget

Slide 8 Status: What's Done, What's Next

1:00

• To summarize where we are: the full pipeline is built, the dataset is acquired, quality-checked, and fully preprocessed, both architectures are implemented and verified, and Experiment 1 is trained, evaluated, and analyzed.

• Still to come for the final review: Experiment 2, the full cross-manipulation generalization matrix; Experiment 3, compression robustness; a Grad-CAM interpretability analysis explaining our results, not just reporting them; and a live working demo.

Slide 9 Closing

0:45

• We're on track, with the harder generalization and robustness questions -- the actual novel contribution of this project -- still to come.

• We look forward to sharing the full comparative results, the interpretability analysis, and a live demo at the final review.

• Thank you -- happy to take questions.